ITT 036 – Programming Assignment 1

Digital Signal Encoder and Decoder

Autumn 2026

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Language: Java **Visualization:** Java Swing and AWT

Objective

Develop a modular Java application that generates digital bit streams, finds the longest palindromic substring, implements digital line encoding and decoding, and demonstrates analog-to-digital conversion with Pulse Code Modulation (PCM) and Delta Modulation (DM).

Implemented Features

- **Digital data generation and palindrome detection:** Generates random binary sequences and sequences containing four or eight consecutive zeros, then identifies the longest palindromic substring.
- **Line coding and decoding:** Implements NRZ-L, NRZ-I, Manchester, Differential Manchester, and Alternate Mark Inversion (AMI), with corresponding decoders.
- **Scrambling:** Supports optional B8ZS and HDB3 scrambling for AMI signals, with corresponding descrambling.
- **Analog-to-digital conversion:** PCM quantizes samples into fixed-width binary codewords and reconstructs approximate samples. Delta Modulation generates a bitstream using a configurable step size and reconstructs a staircase approximation.
- **Visualization:** Displays encoded signal waveforms and reconstructed analog samples using Java graphical components.

Design and Assumptions

The program separates data generation, palindrome detection, line encoders and decoders, scrambling, modulation, and visualization into dedicated classes. Digital signals are represented as arrays of voltage-level samples. PCM uses a configurable number of bits per sample and reconstructs values from quantization intervals. Delta Modulation uses a positive step size; its reconstructed signal is an approximation of the input. Input samples are expected to be normalized to the range [–1, 1].

How to Run in Visual Studio Code

1. Open the project folder in Visual Studio Code and ensure a compatible JDK is installed and available in the terminal.
2. Open the `src` folder in the integrated terminal (Terminal → New Terminal).
3. Compile the project using the command below, then run `java Main`.

```
javac Main.java data\*.java encoding\*.java decoding\*.java scrambling\*.java modulation\*.java visualization\*.java
```

Follow the menus to select digital or analog input, coding/modulation, and optional scrambling.

Testing and Verification

The student reports successful integration tests for the tested configurations, including PCM with NRZ-L, Delta Modulation, and AMI scrambling with B8ZS and HDB3. Standalone known-case tests for B8ZS and HDB3 also passed. These results apply to the tested inputs and do not establish correctness for every possible input.

References

- ITT 036, Programming Assignment 1 specification, Autumn 2026.
- Oracle, Java Platform Documentation: https://docs.oracle.com/en/java/